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React js

Understanding the Virtual DOM in React .

What is the Virtual DOM?

- The Virtual DOM (**VDOM**) is a lightweight, in-memory representation of the real DOM (**Document Object Model**)
- It helps React manage **UI** updates more efficiently by keeping a virtual version of the **UI** in memory
- When changes occur, React updates only the necessary parts of the **real DOM**, instead of re-rendering everything.

1. How Does the Virtual DOM Work?

- **Rendering the Virtual DOM:** React creates a virtual representation of the UI as a tree of JavaScript objects.
- **Updating State:** It generates a new Virtual DOM tree to reflect the updated state when the application state changes.
- **Diffing Algorithm:** React compares the new Virtual DOM tree with the previous one using its efficient diffing algorithm to identify the minimal set of changes required.
- **Updating the Real DOM:** React applies only the necessary changes to the real DOM, optimizing rendering performance.

2. **Pros, Cons, and Use Cases**

Advantages

- **Batch updates → Faster rendering**
- **Cross-browser consistency**
- **Component-friendly updates**
- **Great for dynamic UIs**

Disadvantages

- **Memory overhead**
- **Steeper learning curve**
- **May be overkill for static or simple apps**

3. Real DOM vs Virtual DOM

